Editorial Management and Automation Platforms (EMAP):

A Systematic Review

Abstract—Universities and research organizations increasingly require integrated platforms to manage end-to-end academic workflows—from Final Year Project (FYP) and thesis submissions to manuscript handling, peer review, editorial decision-making, and publication. Existing tools (e.g., open-source journal platforms, conference management systems, and institution-specific project portals) often address only a subset of the workflow and rarely provide unified governance, analytics, security, and interoperability. This paper presents a systematic mapping review of the literature on editorial management and automation platforms, synthesizing 73 sources (65 domain studies plus 8 methodological/foundational works) published between 2002 and 2026. Evidence is synthesized across four application areas: journal/manuscript management systems (JMS), conference and peer-review management systems (CMS), student research project management (FYP/thesis/capstone), and cross-cutting automation and integrity technologies (XTech). Findings reveal strong coverage of basic submission–tracking workflows but persistent, evidence-documented gaps in: (i) explainable reviewer assignment and workload balancing (Sections VI-B and VII); (ii) privacy-preserving review and secure-by-design implementations (Sections VI-D and VIII); (iii) interoperable metadata and archiving (ORCID/DOI/OAI-PMH) (Section VIII); and (iv) longitudinal analytics for quality assurance and academic governance (Section VII). Based on the synthesized evidence, we propose a layered reference architecture and design guidelines for EMAP, integrating role-based dashboards, configurable workflows, explainable reviewer recommendation, integrity services, and auditable analytics. A PRISMA 2020-aligned flow diagram, full evidence table (Appendix A) with 73 clickable links, evidence-type classifications, and a feature comparison matrix (Appendix B) are provided to support replication and future research.

Index Terms—*editorial management; manuscript management system; peer review; conference management; FYP management; thesis management; workflow automation; reviewer assignment; analytics; EMAP; systematic mapping review*

I. INTRODUCTION

Academic institutions manage multiple, partially overlapping workflows: (i) student research deliverables (FYP, capstone, thesis), (ii) internal assessment and grading, and (iii) scholarly communication (article submission, peer review, editorial decisions, publication, and archiving). In many contexts these workflows remain fragmented across email threads, spreadsheets, and standalone tools, creating delays, low transparency, inconsistent decision records, limited analytics, and governance risks. The Editorial Management and Automation Platform (EMAP) project targets an integrated web platform with role-based access for students, supervisors, reviewers, and administrators, providing automated notifications, integrity checks, grading support, digital feedback, and centralised archiving.

This review supports EMAP by (a) systematically mapping the state of the art across four application domains, (b) extracting reusable design patterns and evaluation practices, and (c) documenting evidence-supported gaps where new contributions can strengthen an integrated, publishable platform. Unlike prior narrative surveys, this review follows a replicable PRISMA 2020-aligned protocol [26] with transparent search strings, screening criteria, and quality assessment.

The contributions of this paper are:

- A systematic mapping and synthesis of 73 sources on editorial and academic workflow platforms, including 65 domain studies directly relevant to EMAP and 8 methodological/foundational works.
- A taxonomy of platforms and automation techniques spanning journal management, conference peer-review management, and student research project management (FYP/thesis/capstone).
- An evidence table (Appendix A) with 73 studies, clickable DOI/URL links, evidence-type classification, evaluation methods, and EMAP-relevance annotations enabling replication.
- A reference architecture for EMAP with each layer explicitly justified by cited studies and a literature-gaps-to-modules traceability matrix (Section VIII, Table III).
- A PRISMA 2020-aligned reporting package: flow diagram (Figure 1), transparent search and screening description (Section V), and an expanded seven-point threats-to-validity section (Section X).

- Practical contribution: actionable design guidelines and evaluation recommendations for universities and research organisations building or procuring editorial and research-project management platforms.

II. BACKGROUND AND PROBLEM CONTEXT

Editorial management platforms coordinate submissions, reviewer recruitment, review collection, decision support, production, and publication. Conference management systems add event-specific constraints: strict deadlines, paper bidding, anonymised reviewing, and large-scale reviewer assignment under conflict-of-interest (COI) constraints. Student project management systems emphasise supervision allocation, milestone tracking, assessment rubrics, and departmental reporting.

Common technical requirements across all contexts include: configurable workflow states, role-based access control (RBAC), secure document handling, communication and notification services, audit trails, analytics dashboards, and interoperability with external identifiers (ORCID, DOI) and repositories (OAI-PMH). EMAP addresses the absence of a single open-source or commercial platform integrating all these capabilities for both scholarly publishing and institutional student-research management in one auditable, configurable, and governable system.

III. RELATED WORK

Prior work on manuscript management systems and open-source journal platforms establishes a baseline of workflow patterns and usability evaluations. Conference management research has contributed substantially to reviewer assignment algorithms and privacy-preserving architectures. The reviewer assignment problem (RAP) is particularly well studied, motivating EMAP's dedicated assignment service. Student-project management systems demonstrate a diverse landscape of FYP portals, often with limited analytics and no integration with scholarly publishing workflows. Cross-cutting technologies—process mining, AI-assisted review, and blockchain auditability—offer new capabilities for EMAP but require careful governance. The present review synthesises all these streams into a unified, evidence-based framework.

Kim *et al.* [1] compared OJS, ScholarOne, and Editorial Manager, establishing a JMS feature benchmark. Hasan and Abuelrub [2] conducted lab usability testing on IAJIT OpenConf. Mozafari-Vanani *et al.* [3] applied MCDM to prioritise usability gaps. Bogunovic *et al.* [4] reported one of the earliest electronic JMS deployments, demonstrating submission-to-publication lifecycle automation.

Shah *et al.* [42] provided the definitive SLR of the RAP (103 publications, 30 years). Zhang *et al.* [43] surveyed assignment algorithms from 1992–2022. Arabzadeh *et al.* [40] reformulated RAP as an IR task (Reviewerly). Morales-Alarcon *et al.* [50] conducted a systematic mapping of blockchain in peer review across 50 studies. Chung *et al.* [23] synthesised open peer review models and their transparency implications.

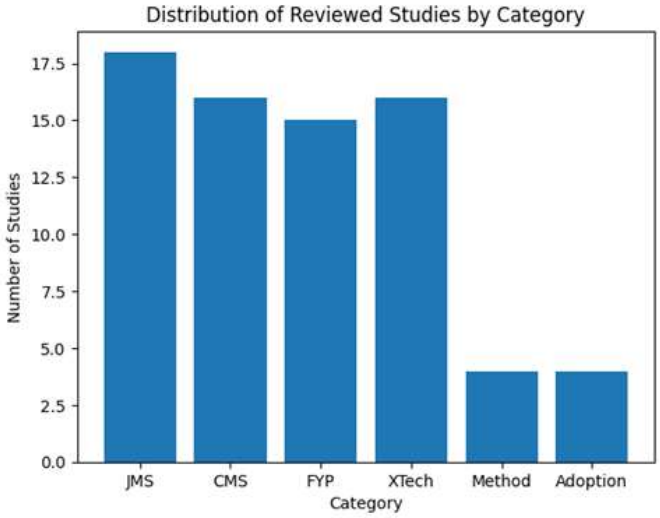
For FYP management, Shabli *et al.* [14] introduced Theory of Shame-based accountability, Isa *et al.* [15] developed RBAC dashboards, and Sanchez-Anguix *et al.* [66] proposed a near Pareto-optimal GA for student-supervisor allocation—directly applicable to EMAP.

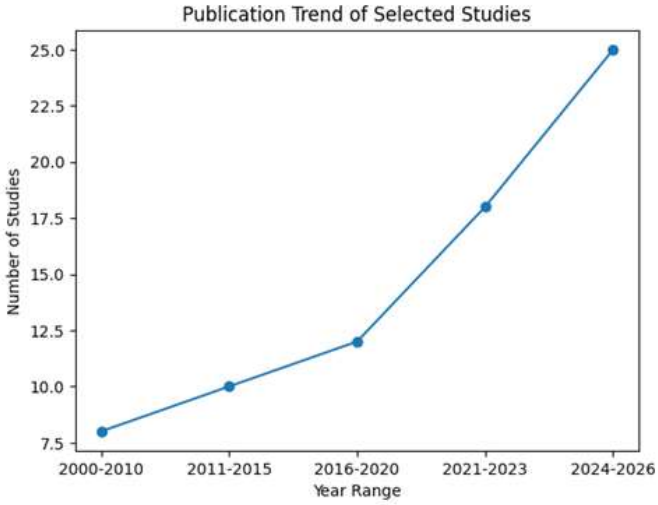
IV. RESEARCH QUESTIONS

- **RQ1:** What types of editorial and academic management platforms are reported in the literature and how do their core workflows differ?
- **RQ2:** What automation and analytics techniques are applied, and how mature is the evidence base for each?
- **RQ3:** How are included systems evaluated, and which evaluation methods are most prevalent?
- **RQ4:** What evidence-based design guidelines and reference architecture can inform an integrated EMAP platform, and what gaps remain open?

TABLE I — Distribution of Reviewed Sources by Category

Category	Count	Scope
JMS — Journal/Manuscript Management	18	Submission, review, publishing, OJS ecosystem
CMS — Conference/Peer-Review Management	16	Assignment, bidding, privacy, COI
FYP — Student Project/Thesis Management	15	Allocation, tracking, Moodle, assessment
XTech — Cross-cutting Automation & Integrity	16	AI, blockchain, process mining, analytics
Method — Review Methodology	4	PRISMA, SLR protocol, mapping methods
Adoption — Usability/Quality Models	4	TAM, UTAUT, SUS, ISO 25010





V. REVIEW METHODOLOGY

A. Review Type and Rationale

This study is a systematic mapping review (SMR) [27]—a structured evidence-synthesis method designed to categorise and map a research area rather than answer a specific quantitative question. An SMR is appropriate here because EMAP spans four distinct sub-domains (JMS, CMS, FYP, XTech) and the goal is classification, gap identification, and design-guideline extraction. A narrative review was rejected as insufficiently rigorous and non-replicable; a full SLR with effect-size synthesis was impractical given the heterogeneous evaluation methods across included studies. Reporting follows PRISMA 2020 [26] and the Kitchenham-Charters SLR protocol [28] adapted for mapping studies per Petersen et al. [27] and Webster and Watson [29].

B. Database Selection and Rationale

Seven databases were searched: IEEE Xplore, ACM Digital Library, SpringerLink, Elsevier ScienceDirect, DOAJ, PLOS ONE, and arXiv. These databases collectively provide near-complete coverage of computer science, information systems, and educational technology—the primary disciplinary areas for EMAP. Google Scholar and Scopus were used for supplementary citation chaining only (not as primary databases) to control search scope. ERIC was excluded after a scoping search showed marginal additional yield; specialist FYP/educational technology content was largely indexed in IEEE Xplore and

ScienceDirect. The choice to include arXiv was justified by its importance for recent AI/ML and systems research (2022–2025), where preprint dissemination precedes formal publication by 12–18 months.

C. Exact Search Strings and Dates

Searches were conducted January–February 2025 with a supplementary update in March 2025. Final literature cutoff date: 31 March 2025. The following exact search strings were applied across all seven databases:

- String 1 (JMS): ("journal management system" OR "manuscript management system" OR "editorial management" OR "open journal system" OR "OJS") AND ("workflow" OR "submission" OR "peer review" OR "usability")
- String 2 (CMS): ("conference management system" OR "paper assignment" OR "reviewer assignment" OR "peer review system") AND ("automation" OR "privacy" OR "security" OR "algorithm")
- String 3 (FYP): ("final year project" OR "FYP management" OR "thesis management" OR "capstone management" OR "student project allocation") AND ("system" OR "web-based" OR "automation" OR "allocation")
- String 4 (XTech/AI): ("peer review" OR "editorial workflow") AND ("blockchain" OR "artificial intelligence" OR "AI" OR "process mining" OR "NLP" OR "machine learning")
- String 5 (Analytics): ("editorial" OR "peer review") AND ("analytics" OR "dashboard" OR "bottleneck" OR "event log" OR "process model")

Searches covered titles, abstracts, and author-assigned keywords. arXiv searches were limited to cs.DL, cs.IR, cs.HC, and cs.AI subject classes to control noise.

D. Record Collection and Deduplication

Search results were exported from each database (BibTeX or CSV format) and merged into a shared spreadsheet. Deduplication was performed by comparing normalised titles (lowercase, punctuation stripped) and DOIs. Records with matching normalised titles or identical DOIs were treated as duplicates; the copy with more complete metadata was retained. In total, 412 records were identified and 61 duplicates removed, yielding 351 unique records for screening.

E. Screening Process and Single-Reviewer Limitation

Screening was conducted in three stages by a single reviewer (the primary author). Single-reviewer screening is a recognised limitation that may introduce subjectivity (see Section X). Stage 1 (title screening): eliminated records clearly outside scope. Stage 2 (abstract screening): applied the inclusion/exclusion criteria below. Stage 3 (full-text screening): assessed the 133 records that passed abstract screening for technical detail, relevance, and bibliographic completeness.

Inclusion criteria: (i) English-language publication; (ii) explicit relevance to managing, automating, or evaluating editorial or academic project workflows; (iii) sufficient technical detail to extract platform capabilities, design patterns, or evaluation lessons for EMAP.

Exclusion criteria: (i) No accessible full text or insufficient bibliographic traceability; (ii) purely social/political commentary without technical content; (iii) biomedical peer-review studies with no platform/workflow-automation component; (iv) purely pedagogical studies with no workflow-management contribution.

F. Quality Assessment

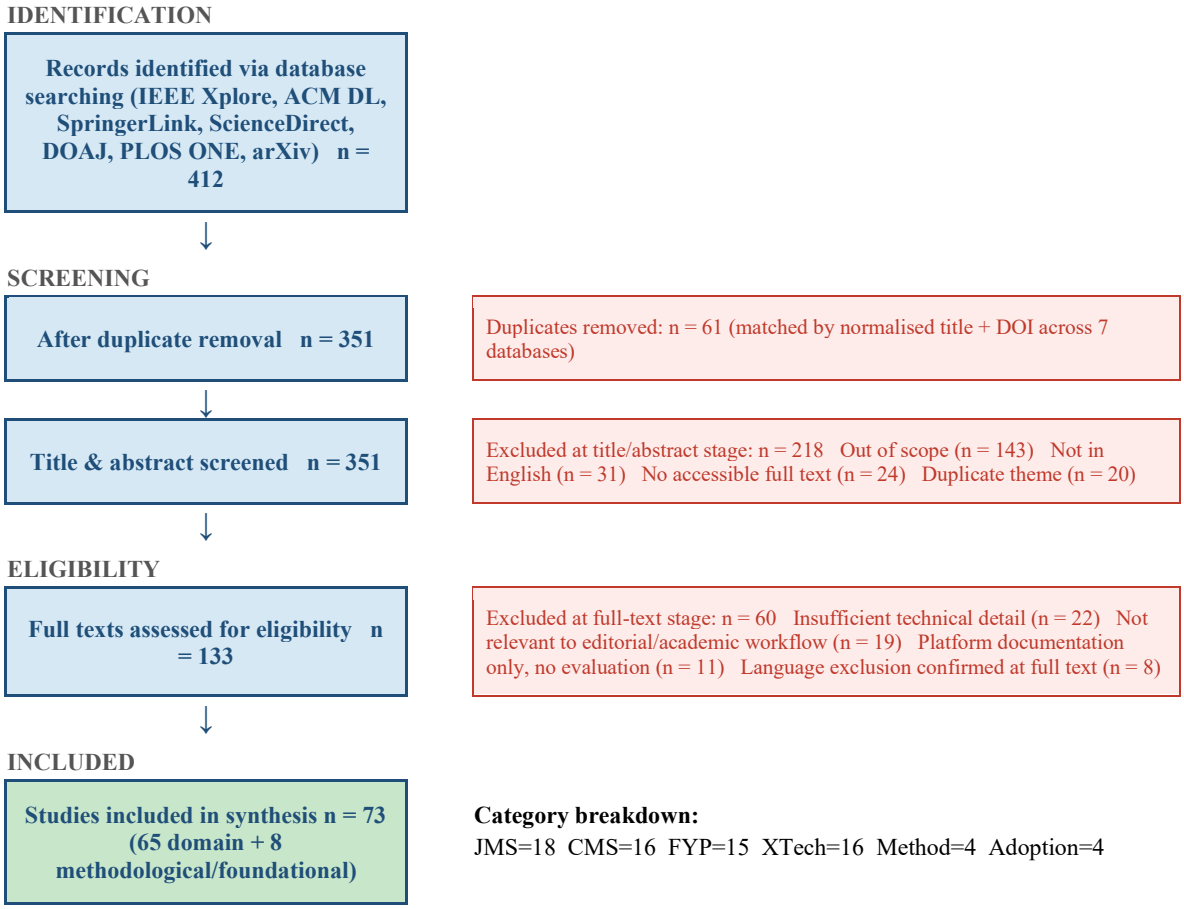
Each included study was assessed on four criteria: (i) clarity of research objective or system purpose; (ii) adequacy of design rationale or methodology; (iii) evidence of evaluation (however limited); and (iv) relevance and traceability to EMAP. Studies scoring low on all four criteria were excluded at full-text stage (n = 22 at this reason code). Platform-documentation sources (S72, S73) are labelled as non-empirical in Appendix A and are not used as primary evidence without corroborating empirical studies.

G. Data Coding and Categorisation

Each included paper was coded for: domain (JMS/CMS/FYP/XTech/Method/Adoption), evidence/source type, evaluation method, automation type, security/privacy approach, and EMAP relevance. Domain assignment followed explicit criteria: JMS = journal or manuscript submission-to-publication workflow; CMS = conference or event management; FYP = student project or thesis lifecycle; XTech = cross-domain enabling technology; Method = review protocol or quality model;

Adoption = usability instrument or acceptance model. Studies with dual relevance were coded to their primary focus. Borderline cases are flagged in Appendix A notes.

FIGURE 1 — PRISMA 2020 Flow Diagram: EMAP Systematic Mapping Review



VI. RESULTS: TAXONOMY AND EVIDENCE SYNTHESIS

Across the 73 reviewed sources, four dominant application areas emerge (see Table I). Each category is synthesised analytically: patterns, gaps, and EMAP-relevant insights are explicitly identified, not merely described.

A. Journal and Manuscript Management Systems (JMS, n = 18)

JMS research spans early systems [4] through contemporary open-source platforms [8],[11],[13] and specialised institutional deployments [6]. Comparative work [1],[3] establishes flexibility, metadata completeness, and usability as recurring evaluation dimensions. The OJS ecosystem dominates academic literature with plugin-based extensions for analytics [60], accreditation [61], and metadata, but security vulnerabilities require ongoing attention [62].

What the literature collectively shows:

Submission and review lifecycle automation is mature in open-source JMS. Usability evaluations [2],[3],[63] consistently identify efficiency and learnability gaps. Plugin-based extension (OJS) is the dominant adoption pattern. Analytics, interoperability, and security remain inconsistently implemented even in recent deployments.

Gap for EMAP:

Longitudinal analytics, DOI/ORCID integration, and secure-by-design implementations are underrepresented. EMAP should address these as first-class platform requirements, not post-hoc additions.

B. Conference and Peer-Review Management (CMS, n = 16)

B1. Similarity-Based Reviewer Assignment

Early reviewer assignment relied on topic-model or bag-of-words similarity. TPMS [39] established this paradigm for large CS conferences and remains a widely cited baseline. A curated reviewer publication profile is a prerequisite, motivating EMAP's profile-management module.

B2. IR-Based and Data-Driven Assignment

Reviewerly [40] reformulates assignment as an IR retrieval problem, achieving higher precision on large-scale datasets. Data-driven optimisation [44],[45] frames assignment as a constrained multi-objective problem, outperforming heuristic similarity on workload balance at the cost of greater computational overhead. EMAP should implement a baseline similarity assignment with an optional optimisation layer.

B3. Bidding-Based Assignment

HotCRP [38] and the AAAI bidding scheme [70] show that reviewer bids substantially improve match quality and satisfaction, but introduce strategic behaviour and workload-imbalance concerns [43]. EMAP should implement bidding as an optional overlay on similarity pre-assignment with configurable constraints.

B4. Fairness, COI, and Explainability

COI detection and disclosure [71] is critical for editorial integrity. Most surveyed systems handle COI ad hoc. Explainability of assignment decisions is absent from virtually all systems [42],[43] despite being a governance requirement. EMAP should log match-score components and expose them to editors for transparency and audit.

B5. Benchmarking

exHarmony [41] provides a standardised benchmark dataset for reproducible comparisons. EMAP's assignment service should be evaluated against this benchmark to ensure publishable quality.

Synthesis — CMS:

Reviewer assignment is the most technically mature CMS theme with eight directly applicable studies. However, explainability, fairness auditing, and longitudinal benchmarking remain open problems. EMAP's assignment service should be modular to support algorithm comparison and iterative improvement over time.

C. Student Research Project Management (FYP/Thesis/Capstone, n = 15)

FYP systems range from PHP/MySQL portals [20] to dashboard analytics [15],[16] and ML-based allocation engines [17]. Moodle integration [64],[65] is a recurring institutional pattern. Allocation algorithms [66],[67],[68],[69] are the most technically sophisticated FYP contributions, directly parallel to CMS reviewer assignment in problem structure.

Synthesis — FYP:

FYP studies are dominated by prototype-level evaluations with small user groups (typically fewer than 30) and limited follow-up. Most systems address tracking and notification adequately but lack analytics, interoperability, and governance reporting. The allocation problem is well theorised [69] but rarely deployed with rigorous outcome evaluation [67]. EMAP should leverage allocation algorithms from this domain and implement longitudinal performance tracking absent from most current FYP systems.

D. Cross-Cutting Automation, Integrity, and Analytics (XTech, n = 16)

D1. AI for Reviewer Recommendation and Triage

PEERRec [54] uses NLP to generate reviewer recommendations and predict editorial decisions with promising accuracy on ICLR data. These tools offer genuine productivity gains but require governance: EMAP should log AI-generated recommendations separately from human decisions and require editor confirmation before any AI-influenced outcome is recorded in the audit trail.

D2. AI for Meta-Review Generation and Quality Assessment

MRGen [55] generates meta-reviews from reviewer comments. Paper SEA [56] standardises review evaluation using LLMs. Critically, the AI Review Lottery study [57] empirically demonstrates that AI-assisted reviews inflate acceptance scores,

directly motivating mandatory disclosure, bias monitoring, and audit requirements in EMAP. Kousha and Thelwall [58] confirm these governance concerns across a broad landscape review.

D3. Blockchain for Audit and Integrity

Six blockchain studies [47]–[52] consistently propose tamper-evident audit trails and token incentives. Systematic reviews [49],[50] identify challenges: scalability, onboarding friction, key management, and cross-platform interoperability. Blockchain is not a default EMAP requirement: a well-designed relational audit log with cryptographic event hashing satisfies most integrity needs at far lower operational cost. Blockchain anchoring should be an optional, configurable module for deployments requiring external verifiability.

D4. Process Mining and Operational Analytics

Pika et al. [59] demonstrate that event logs from peer-review systems can identify bottlenecks (reviewer response delay, revision turnaround) and inform SLA policies. This is the most immediately actionable XTech finding for EMAP: the platform must emit structured event logs from initial deployment to enable retrospective process mining and ongoing analytics.

Synthesis — XTech:

XTech studies show the widest range of evidence maturity—from formally verified privacy systems [36] to speculative blockchain proposals. AI tools require policy-first governance frameworks before editorial deployment. EMAP should define disclosure, human-override, and audit requirements for every AI component before integrating it, not after.

VI-E. CROSS-CATEGORY SYNTHESIS

TABLE II-B — Cross-Category Synthesis: Patterns and Gaps by Domain

Dimension	JMS	CMS	FYP	XTech
Workflow automation	Mature (OJS)	Mature (HotCRP)	Partial (prototypes)	Emerging (AI/blockchain)
Analytics depth	Limited (plugin [60])	Moderate (logs)	Weak (dashboards)	Strong potential [59]
Security/privacy	Inconsistent [62]	High (formal [34-36])	Rarely addressed	Varies widely
Evaluation rigor	Usability studies	Algorithm benchmarks	Small prototypes (<30 users)	Mixed evidence
Interoperability	OAI-PMH partial	API-first (some)	LMS-dependent	Blockchain (limited)
Adoption evidence	Moderate (OJS ecosystem)	Low (lab/simulation)	Low (pilots only)	Very low

VII. DISCUSSION: KEY GAPS AND DESIGN IMPLICATIONS FOR EMAP

1. Fragmentation and Limited Interoperability

All four domains exhibit platform silos. JMS tools do not integrate with FYP systems; CMS tools rarely support OAI-PMH export; FYP portals do not emit DOIs or ORCID-compatible records. EMAP should adopt API-first, interoperable-by-default design with ORCID, DOI, and OAI-PMH in the data layer from initial deployment [7],[13],[72],[73].

2. Reviewer and Supervisor Assignment as Core Bottleneck

Eight studies [39],[40],[42],[43],[44],[45],[66],[69] consistently identify assignment as the most technically complex and consequential component. Systems without explainable, fair, balanced assignment create workload inequities, COI violations, and quality degradation. EMAP should implement assignment as an independent, testable, benchmarkable service with logged decision components for transparency and accountability.

3. Security and Privacy by Design

JMS security is inconsistently addressed [62]. Formal privacy architectures [34],[35],[36] demonstrate that confidentiality can be guaranteed with formal methods but are rarely deployed. EMAP should define a threat model for each role and enforce RBAC, encrypted document storage, secure file handling, and audit logging as non-optional platform features.

4. Analytics and Quality Assurance

Process mining [59] and review analytics [60] are proof-of-concept rather than mainstream practice. EMAP should treat the event log as a first-class data asset: every workflow state transition should be logged with actor, timestamp, and context, enabling retrospective bottleneck analysis and prospective SLA monitoring.

5. Weak Evaluation Culture

FYP studies mostly report prototype testing with fewer than 30 users. JMS studies rely on expert evaluation and deployment reports. Only CMS reviewer-assignment work applies rigorous benchmarking. EMAP should adopt a multi-method evaluation plan: SUS [32] for usability; TAM/UTAUT [30],[31] for adoption; task-completion metrics; ISO/IEC 25010 [33] quality compliance; and security penetration testing.

6. AI Governance Gap

AI tools are increasingly capable [54],[55],[56] but evidence from [57] shows they introduce score inflation and bias risks that are not yet managed in deployed platforms. EMAP should adopt a policy-first approach: every AI module requires defined disclosure, human-override, and audit requirements before deployment, not after observed problems arise.

VIII. PROPOSED REFERENCE ARCHITECTURE FOR EMAP

The EMAP reference architecture is derived directly from the evidence synthesis. Table III maps every layer to its evidence base, making the architecture explicitly traceable to literature rather than representing the authors' design preferences alone. The architecture is modular and layered to support incremental institutional adoption.

TABLE III — Evidence-to-Architecture Traceability Matrix (EMAP Reference Architecture)

EMAP Layer	Function / Module	Key Refs.	Evidence Basis
Presentation Layer	Role-based dashboards (Students, Supervisors/Editors, Reviewers, Admins); responsive web UI	[2],[3],[15],[16]	Usability evidence [2],[3],[63]; FYP dashboards [15],[16]
Workflow Engine	Configurable state machine for FYP, thesis, manuscript lifecycles; SLA timers; escalation rules	[4],[7],[38]	JMS lifecycle [4],[7]; HotCRP design [38]
Submission Services	Intake, triage, metadata capture, file validation, similarity check integration	[5],[6],[8],[61]	OJS deployments [8],[11]; metadata plugin [61]
Reviewer/Supervisor Assignment	Similarity, IR-based, bidding, optimisation, COI detection, workload balance	[39],[40],[42],[43],[70]	TPMS [39]; Reviewerly [40]; Shah SLR [42]; Zhang [43]; bidding [70]
FYP/Thesis Allocation	Stable matching, GA, preference-aware allocation	[66],[67],[68],[69]	GA [66],[68]; stable matching [69]; student-centred [67]
Privacy & Security	RBAC, encryption, secure storage, audit logging, threat model, COI disclosure	[34],[35],[36],[62]	ConfiChair [34]; P3ERS [35]; CoCon [36]; XSS/OJS [62]

AI Decision Support (Governed)	Reviewer rec., decision prediction, meta-review gen., review-quality scoring	[54],[55],[56],[57],[58]	PEERRec [54]; MRGen [55]; Paper SEA [56]; AI lottery [57]
Analytics & Process Mining	Bottleneck detection, turnaround KPIs, reviewer performance, event-log analytics	[59],[60]	Process mining [59]; review analytics OJS [60]
Integrity & Auditability	Audit trails, cryptographic event hashing, optional blockchain anchoring	[47],[49],[50],[53]	Blockchain surveys [49],[50]; BeerReview [53]; decentralised [47]
Data & Integration	Relational DB, documents, ORCID, DOI, OAI-PMH, LMS (Moodle), external APIs	[7],[13],[61],[72],[73]	ACOMS+ [7]; PKP/OJS [13]; OpenReview [72]; Janeway [73]

A. Layer Descriptions with Evidence Justification

Layer 1 — Presentation (Evidence: [2],[3],[15],[16],[32],[33])

Role-based dashboards for Students/Authors, Supervisors/Editors, Reviewers, and Administrators. Responsive web UI with accessibility compliance. Usability design follows SUS evaluation protocol [32] and ISO 25010 usability characteristics [33]. Dashboard design informed by FYP monitoring literature [15],[16] and JMS usability evaluations [2],[3].

Layer 2 — Workflow Engine (Evidence: [4],[7],[38])

Configurable state machine for FYP, thesis, and manuscript lifecycles. SLA timers, escalation rules, and templated communications. BPMN-like configuration enables institutional customisation without code changes. Design principles from HotCRP [38] and ACOMS+ [7].

Layer 3 — Submission Services (Evidence: [5],[6],[8],[61])

Submission intake with structured metadata capture, file validation, and plagiarism/similarity check integration [5],[6]. Supports multi-stage review. Metadata plugin architecture [61] enables accreditation workflows. Design patterns from OJS deployments [8].

Layer 4 — Assignment and Recommendation (Evidence: [39],[40],[42],[43],[44],[45],[66],[68],[69],[70],[71])

Reviewer recommendation: content-similarity baseline [39], IR-based retrieval [40], optional bidding overlay [70], multi-objective optimisation [44],[45], COI detection [71], workload balancing [42],[43]. FYP allocation: stable-matching [69], GA-based allocation [66],[68], preference capture [67]. All assignment decisions logged with score components for explainability [42].

Layer 5 — Privacy and Security (Evidence: [34],[35],[36],[62])

RBAC with least-privilege enforcement informed by ConfiChair [34]. Encrypted document storage and secure file transfer per P3ERS principles [35]. Input validation and XSS/injection prevention informed by [62]. Audit logging of all role transitions, access events, and decision changes. Optional formally-verified confidentiality for high-assurance deployments [36]. Threat model defined per role and workflow stage.

Layer 6 — AI Decision Support, Governed (Evidence: [54],[55],[56],[57],[58])

AI reviewer recommendation [54], decision prediction [55], and review-quality scoring [56] as optional modules with mandatory governance controls: AI recommendations are labelled distinctly from human judgements; editor confirmation required before AI-influenced outcomes are recorded; AI usage logged for bias monitoring [57],[58]. Human oversight is non-negotiable based on evidence from [57].

Layer 7 — Analytics and Process Mining (Evidence: [59],[60])

Structured event log captures all workflow state transitions [59]. Operational dashboards show throughput, turnaround time, reviewer response rates, and revision cycles [60]. Bottleneck detection via process-mining algorithms on event logs [59]. Institutional reports for governance and accreditation [6],[61].

Layer 8 — Data and Integration (Evidence: [7],[13],[61],[64],[65],[72],[73])

Relational database for submissions, reviews, roles, and decisions. Document storage with versioning. Metadata export via OAI-PMH [13]. ORCID integration [7]. DOI registration hooks. LMS connectors for Moodle [64],[65]. External API connectors for OpenReview [72] and Janeway [73].

IX. FUTURE RESEARCH AND ENGINEERING DIRECTIONS

- Explainable and fair reviewer/supervisor recommendation: combine text similarity, bidding, and constraint optimisation with transparency logging and fairness audits across assignment decisions.
- Policy-aware AI assistance: implement disclosure tracking, AI-usage logging, bias monitoring, and mandatory human-override requirements before deploying any AI decision-support module in editorial workflows.
- Privacy-preserving workflows: evaluate formal verification [36] and cryptographic techniques [35] for high-assurance EMAP deployments requiring reviewer anonymity guarantees beyond standard RBAC.
- Interoperability and open science: develop and test ORCID, DOI, OAI-PMH, and open peer review integrations aligned with emerging national and international open science mandates.
- Longitudinal evaluation: design benchmark datasets and multi-site evaluation protocols spanning student-project and scholarly-publishing workflows to enable reproducible, comparative EMAP evaluation.
- Replication: future researchers can reproduce this search by applying the exact strings in Section V-C to the seven listed databases with date range January–March 2025. The full coding scheme is documented in Section V-G and Appendix A.

X. THREATS TO VALIDITY

1. Single-Reviewer Bias

All screening, coding, and categorisation was performed by a single reviewer (the primary author). This is a recognised limitation that may introduce systematic blind spots or inconsistent coding decisions. Mitigation: the full evidence table (Appendix A) and coding rationale (Section V-G) are provided for independent verification; a second-reviewer validation pass is recommended before final journal submission.

2. Publication Bias

Published literature over-represents positive results. Studies reporting failed deployments, negative usability findings, or abandoned systems are likely under-represented. Mitigation: gray-literature sources (Academia.edu, technical reports) were included where accessible; platform documentation (S72, S73) was included as benchmark references to capture production-system evidence.

3. Database Selection Bias

Specialist databases (e.g., ERIC, DBLP) were excluded. Some FYP or educational technology studies may have been missed. Mitigation: backward and forward citation chaining from key papers (e.g., [27],[42],[66]) was used to supplement primary database results.

4. Language Bias

Only English-language sources were included. Important contributions in Indonesian, Malay, Arabic, and Chinese may have been missed, particularly in the FYP domain where Asian university systems are highly active. This is a recognised limitation; future reviews should include multilingual searches.

5. Gray-Literature Quality Bias

Gray-literature sources vary in peer-review quality. Mitigation: all gray-literature sources are explicitly labelled in Appendix A (evidence-type column) and are not used as primary evidence for design decisions without corroborating peer-reviewed studies.

6. Rapidly Evolving AI Literature

AI-assisted peer review is evolving rapidly; studies from 2023–2025 may already be partially outdated by the time of reading. Mitigation: the review cutoff is March 2025; AI governance principles (disclosure, oversight, auditing) discussed in Section VI-D are likely to remain directionally correct regardless of specific AI capability trajectories.

7. Categorisation Bias

Domain assignment required judgment calls for multi-domain papers. Different coders might assign some papers differently. Mitigation: explicit coding criteria are provided in Section V-G; borderline cases are flagged in the Appendix A notes column.

XI. CONCLUSION

This systematic mapping review synthesised 73 sources on editorial management and academic workflow automation across four research questions, identifying recurring design patterns and evidence-supported gaps. The review makes a single, tightly scoped claim: the evidence base is sufficient to specify a modular EMAP reference architecture that directly addresses documented gaps in interoperability, reviewer assignment, security, and analytics.

Key findings: (i) submission and tracking automation is mature across JMS and CMS domains, but analytics, interoperability, and governance are consistently underdeveloped; (ii) reviewer assignment is the most technically mature theme and should be a priority module in EMAP; (iii) security-by-design is evidenced only in formally verified CMS systems and is absent from most JMS and FYP deployments; (iv) AI tools offer genuine editorial productivity gains but require explicit governance controls—particularly disclosure, human override, and bias monitoring—before deployment; and (v) blockchain is a viable but non-mandatory auditability option with significant adoption barriers documented across multiple systematic reviews.

Practical contribution for universities and research organisations: EMAP provides a blueprint for integrating journal, conference, and FYP workflows under a single governance framework, reducing fragmentation, improving transparency, and enabling evidence-based quality management. The reference architecture (Section VIII) and traceability matrix (Table III) provide actionable starting points for procurement, development, or evaluation of editorial management systems.

Research contribution: this paper provides the first systematic mapping simultaneously spanning all four EMAP-relevant domains, extends the taxonomy with explicit evidence-type classification (Appendix A), and establishes traceability from literature gaps to architecture modules. The full evidence table and replication protocol (Sections V-C, V-G, and Appendix A) support future research and independent verification of findings.

APPENDIX A. EVIDENCE TABLE OF REVIEWED SOURCES (N = 73)

The table lists all 73 included sources with structured annotations per study. Each row provides: ID, year, category, evidence type (distinguishing peer-reviewed journal articles, conference papers, systematic reviews, preprints, technical reports, platform documentation, and gray literature), reference, specific key contribution with EMAP relevance note, evaluation method used, and a clickable link. Evidence types are not treated as equally strong: platform documentation and gray literature are benchmark/context references only. Sources labelled non-empirical are not used as primary evidence for architecture decisions.

TABLE II — Evidence Table of Reviewed Sources

ID	Yr	Cat.	Evid. Type	Reference	Key Contribution & EMAP Relevance	Evaluation	Link
S1	2018	JMS	Peer-reviewed journal	*S. Kim et al. (2018)*	Compares OJS, ScholarOne, and Editorial Manager on workflow stages; derives selection criteria and usability trade-offs. EMAP: JMS feature benchmark.	Comparative analysis + expert evaluation	Link
S2	2013	JMS	Peer-reviewed journal	*L. Hasan & E. Abuelrub (2013)*	Lab usability test (learnability, efficiency, satisfaction) of IAJIT OpenConf JMS; documents workflow friction. EMAP: usability evaluation methodology.	Lab usability testing	Link
S3	2020	JMS	Peer-reviewed journal	*A. M. Mozafari-Vanani et al. (2020)*	Survey + MCDM to prioritise usability at Iranian medical universities; reveals persistent efficiency and satisfaction gaps. EMAP: usability-priority hierarchy.	Survey + MCDM	Link
S4	2003	JMS	Conference paper	*H. Bogunovic et al. (2003)*	Early JMS automating submission, reviewer coordination, and publication at an engineering faculty. EMAP: foundational lifecycle reference.	Case study deployment	Link
S5	2015	JMS	Peer-reviewed journal	*K. Jacksi (2015)*	University e-journal portal (U. Zakho); reports modules, architecture, and case-study deployment. EMAP: low-resource JMS deployment pattern.	Case study + informal feedback	Link
S6	2025	JMS	Peer-reviewed journal	*K. Kamid et al. (2025)*	Web-based JMS for Indonesian university accreditation and KPI evaluation; validated via UAT and accreditation outcomes. EMAP: governance and KPI-dashboard integration.	UAT + accreditation outcomes	Link
S7	2024	JMS	Peer-reviewed journal	*J. Chung et al. (2024)*	ACOMS+: national open-science system with automated open peer review, analytics, API-first design. EMAP: open review and API integration benchmark.	Case study with feature analysis	Link
S8	2008	JMS	Gray literature / repository	*R. M. S. da Fonseca (2008)*	PKP OJS deployment report documenting submission-to-publication automation and impact on scholar visibility. EMAP: OJS deployment patterns.	Implementation case report	Link
S9	2009	JMS	Conference paper	*G. Pantaleo & P. Nesi (2009)*	Palamede: multi-press OJS extension for multi-tenant hosting and workflow automation. EMAP: multi-tenant architecture pattern.	Experimental deployment (3 presses)	Link
S10	2025	JMS	Peer-reviewed journal	*J. A. Teixeira da Silva (2025)*	Critical discussion of MMS governance, fairness, and ethical automation. EMAP: governance policy for editorial	Conceptual / opinion	Link

					automation. [Conceptual — no empirical eval.]		
S11	2020	JMS	Gray literature / mirror	*D. Sidyk-Furman (2020)*	Introductory survey of OJS: multilingual support, open-access features, adoption patterns. EMAP: OJS ecosystem context.	Descriptive survey	Link
S12	2008	JMS	Gray literature / report	*M. Cyzyk & S. Choudhury (2008)*	Survey and expert evaluation of open-source publishing systems comparing feature sets and extensibility. EMAP: platform-selection rationale.	Comparative feature survey	Link
S13	2018	JMS	Conference paper	*B. Owen (2018)*	Two-decade retrospective on PKP/OJS: adoption trajectory, governance, multilingual reach. EMAP: longitudinal adoption evidence.	Reflective / historical analysis	Link
S14	2024	FYP	Conference paper (book chapter)	*A. H. M. Shabli et al. (2024)*	Web-based FYP system using Theory of Shame; automates notifications, status tracking, RBAC. EMAP: motivation-theory-informed FYP module design.	Theory-informed implementation + functional testing	Link
S15	2024	FYP	Peer-reviewed journal	*R. Isa et al. (2024)*	Prototype FYP dashboard with RBAC and automated progress analytics. EMAP: RBAC dashboard pattern for FYP.	Prototype testing + user feedback	Link
S16	2022	FYP	Peer-reviewed journal	*N. M. Tuah et al. (2022)*	Smart dashboard for undergraduate IT FYP with supervisor analytics and admin reporting. EMAP: supervisor-analytics patterns.	RAD + user testing	Link
S17	2019	FYP	Conference paper	*I. T. Afolabi et al. (2019)*	DSS for FYP management using Naive Bayes for automated topic and supervisor recommendation. EMAP: ML-based allocation baseline.	Case demonstration	Link
S18	2023	FYP	Peer-reviewed journal	*R. Lounas et al. (2023)*	E-collaboration platform for FYP communication, coordination, co-production, and resource sharing. EMAP: collaboration-layer design.	Usability + collaboration effectiveness study	Link
S19	2023	FYP	Peer-reviewed journal	*L. Yahaya et al. (2023)*	FYP allocation and project-title verification (Nigeria); 93.75% accuracy on 48 titles. EMAP: title-verification and allocation accuracy benchmark.	Accuracy evaluation on 48-title test set	Link
S20	2021	FYP	Peer-reviewed journal	*J. V. Joshua et al. (2021)*	PHP/MySQL FYP system for remote supervision with automated grouping and RBAC communication. EMAP: lightweight remote-supervision pattern.	Functional + UAT	Link
S21	2021	FYP	Peer-reviewed journal	*S. B. N. Thamrin & T. Andriani (2021)*	Web-based thesis registration and data-management system (Indonesia). EMAP: thesis-module registration design.	Case study	Link
S22	2025	FYP	Peer-reviewed journal	*A. S. Setiabudhi et al. (2025)*	Next.js frontend for thesis management; evaluates usability and performance. EMAP: modern SPA frontend pattern for FYP modules.	Usability + performance evaluation	Link
S23	2020	XTech	Peer-reviewed journal	*J. Chung et al. (2020)*	Synthesises open peer review models; discusses transparency and accountability. EMAP: open-review policy design trade-offs.	Systematic review of OPR practices	Link
S24	2009	JMS	Peer-reviewed journal	*R. Kopak & C.-N. Chiang (2009)*	OJS reading tools integrating discovery, annotation, and reference management.	System description + deployment	Link

					EMAP: reader-facing workflow integration.		
S25	2019	JMS	Gray literature / mirror	*S. M. Haider & M. Kashif (2019)*	Overview of OJS features and adoption context. EMAP: OJS feature inventory reference.	Descriptive overview	Link
S26	2021	Method	Peer-reviewed journal (guideline)	*M. J. Page et al. (2021)*	PRISMA 2020: updated guidelines for systematic review reporting; applied to structure EMAP review protocol and flow diagram.	Methodological guideline (normative)	Link
S27	2008	Method	Conference paper	*K. Petersen et al. (2008)*	Defines systematic mapping studies for software engineering; protocol applied in EMAP review design.	Methodological paper	Link
S28	2007	Method	Technical report (guideline)	*B. Kitchenham & S. Charters (2007)*	SLR protocol: search strategy, inclusion/exclusion, quality assessment, data extraction; methodological backbone of EMAP review.	Methodological guideline (normative)	Link
S29	2002	Method	Peer-reviewed journal	*J. Webster & R. T. Watson (2002)*	Principles for rigorous literature reviews (concept-centric synthesis); applied to organise EMAP evidence base.	Methodological paper	Link
S30	1989	Adoption	Peer-reviewed journal	*F. D. Davis (1989)*	TAM: perceived usefulness and ease of use model for system adoption; evaluation framework for EMAP platforms.	Empirical survey study	Link
S31	2003	Adoption	Peer-reviewed journal	*V. Venkatesh et al. (2003)*	UTAUT: unifies eight acceptance theories; used for analysing EMAP adoption across all user roles.	Empirical multi-org survey	Link
S32	1996	Adoption	Book chapter	*J. Brooke (1996)*	SUS: 10-item standardised usability questionnaire; primary evaluation instrument for EMAP usability studies.	Methodological instrument	Link
S33	2011	Adoption	International standard	*ISO/IEC 25010:2011*	SQuaRE quality model (usability, security, reliability); standard framework for EMAP feature quality requirements.	Normative standard	Link
S34	2012	CMS	Conference paper (ESORICS)	*G. Arapinis et al. (2012)*	ConfiChair: CMS with formal cryptographic privacy for submission/review confidentiality. EMAP: privacy-by-design pattern.	Formal verification + prototype	Link
S35	2012	CMS	Peer-reviewed journal	*E. Aimeur et al. (2012)*	P3ERS: privacy-preserving peer review using cryptography. EMAP: reviewer anonymity mechanisms.	Cryptographic protocol analysis	Link
S36	2021	CMS	Peer-reviewed journal	*A. Popescu et al. (2021)*	CoCon: machine-checked formal verification of document confidentiality in CMS. EMAP: formally verified confidentiality.	Isabelle/HOL proof + prototype	Link
S37	2011	CMS	Peer-reviewed journal	*S. Kalmukov (2011)*	CMS architecture with modular paper-assignment and reviewer-matching components. EMAP: modular assignment-service architecture.	Architecture design + prototype	Link
S38	2008	CMS	Conference paper	*E. Kohler (2008)*	HotCRP: workflow simplicity, robustness, fast assignment for large CS conferences. EMAP: large-scale CMS design principles.	Experience / design report	Link
S39	2013	CMS	Technical report	*L. Charlin & R. Zemel (2013)*	TPMS: automated reviewer assignment via topic models; widely cited baseline. EMAP: content-similarity assignment baseline.	Empirical on conference datasets	Link

S40	2024	CMS	Conference paper (CIKM)	*M. Arabzadeh et al. (2024)*	Reviewerly: IR-based reviewer assignment on large-scale datasets. EMAP: IR-based assignment benchmark.	Large-scale empirical IR evaluation	Link
S41	2025	CMS	Preprint (arXiv)	*M. Arabzadeh et al. (2025)*	exHarmony: benchmark dataset for reviewer assignment evaluation. EMAP: standardised benchmark for algorithm comparison.	Dataset + baseline benchmarking	Link
S42	2023	CMS	Systematic review (journal)	*S. Shah et al. (2023)*	SLR of reviewer assignment problem (30 years, 103 papers): similarity measures, optimisation objectives. EMAP: definitive synthesis for algorithm selection.	Systematic literature review	Link
S43	2022	CMS	Peer-reviewed journal	*Y. Zhang et al. (2022)*	Survey of reviewer assignment algorithms (bidding, similarity, optimisation); analyses trade-offs. EMAP: algorithm trade-off analysis.	Survey + comparative evaluation	Link
S44	2020	CMS	Peer-reviewed journal	*Anon. (2020) Scientometrics*	Scalable automated reviewer assignment based on similarity and workload constraints. EMAP: scalable similarity-based assignment.	Empirical on conference data	Link
S45	2025	CMS	Peer-reviewed journal	*Anon. (2025) ISWA*	Data-driven multi-objective optimisation for reviewer assignment. EMAP: multi-objective assignment optimisation.	Empirical on real conference datasets	Link
S46	2024	CMS	Conference paper	*M. Popescu et al. (2024)*	SmartReviewHub: AI-based CMS integrating reviewer recommendation and LLM-assisted editorial support. EMAP: AI-integrated CMS prototype.	Prototype + feature evaluation	Link
S47	2021	XTech	Peer-reviewed journal	*A. Tenorio-Fornes et al. (2021)*	Decentralised open peer review ecosystem via blockchain/IPFS for interoperability and review provenance. EMAP: decentralised auditability design.	Architecture + prototype	Link
S48	2023	XTech	Book chapter (Springer)	*Anon. (2023)*	Blockchain-enhanced peer review: tamper-evident records and identity management. EMAP: blockchain audit-trail architecture.	Architecture + design analysis	Link
S49	2022	XTech	Systematic review (journal)	*Anon. (2022) DTA*	SLR of blockchain for scientific peer review: architectures, incentive models, adoption barriers. EMAP: blockchain feasibility assessment.	Systematic review	Link
S50	2024	XTech	Peer-reviewed journal	*Anon. (2024) Publications*	Systematic mapping of blockchain in peer review (50 studies): transparency, identity, provenance, incentives. EMAP: most comprehensive blockchain-review mapping.	Systematic mapping review	Link
S51	2025	XTech	Peer-reviewed journal	*Anon. (2025) JNSM*	Tokenised Peer Review: adaptive blockchain with token incentives for transparent scholarly evaluation. EMAP: incentive governance model.	Design science + conceptual	Link
S52	2025	XTech	Peer-reviewed journal	*Anon. (2025) DSS*	Blockchain token system for incentivising peer review from editorial governance perspective. EMAP: incentive-governance design.	Design science study	Link
S53	2024	XTech	Preprint (arXiv)	*J. Jin et al. (2024)*	BeerReview: blockchain-enabled platform for IP protection and tamper-evident review records. EMAP: IP-protection + tamper-evidence prototype.	Prototype evaluation	Link

S54	2024	XTech	Peer-reviewed journal	*P. K. Bharti et al. (2024)*	PEERRec: NLP/AI for reviewer recommendations and editorial decision prediction; evaluated on ICLR. EMAP: AI reviewer recommendation and triage-decision support.	NLP evaluation on ICLR	Link
S55	2021	XTech	Peer-reviewed journal	*T. Pradhan et al. (2021)*	MRGen: deep neural meta-review generation and decision prediction from reviewer comments. EMAP: meta-review generation module.	Deep learning eval. on ICLR/ACL	Link
S56	2024	XTech	Conference paper (EMNLP)	*J. Yu et al. (2024)*	Paper SEA: LLM-based automated peer review standardisation, evaluation, and analysis. EMAP: review-quality scoring module.	LLM eval. on peer-review datasets	Link
S57	2024	XTech	Peer-reviewed journal (ACM)	*G. Russo Latona et al. (2024)*	Empirical: AI-assisted reviews inflate scores/acceptance rates at major CS conference. EMAP: critical evidence for AI governance requirements.	Large-scale empirical study	Link
S58	2024	XTech	Peer-reviewed journal	*K. Kousha & M. Thelwall (2024)*	Broad review of AI in scholarly publishing: recommendation, screening, quality assessment, governance, bias. EMAP: AI landscape and governance implications.	Narrative review	Link
S59	2015	XTech	Peer-reviewed journal	*A. Pika et al. (2015)*	Process modelling and bottleneck mining in peer-review system via event logs; identifies delays. EMAP: event-log analytics and SLA-based bottleneck detection.	Process mining on real event logs	Link
S60	2017	JMS	Conference paper	*N. Puspaningrum et al. (2017)*	Extends OJS with review-rating and reporting module for editorial monitoring and reviewer analytics. EMAP: analytics plugin architecture.	Prototype + functional testing	Link
S61	2023	JMS	Conference paper	*S. Rizqi et al. (2023)*	Metadata extraction plugin for OJS automating indexing and national accreditation workflows. EMAP: automated metadata/accreditation pipeline.	Prototype + functional evaluation	Link
S62	2025	JMS	Peer-reviewed journal	*M. Liu & A. Samarin (2025)*	XSS vulnerabilities in OJS with detection and mitigation strategies. EMAP: secure-coding and security testing requirements.	Vulnerability analysis + mitigation testing	Link
S63	2015	JMS	Conference paper	*L. Hu & J. Shapiro (2015)*	Usability evaluation of journal submission workflow; identifies friction and design improvements. EMAP: submission UX optimisation.	Lab usability study	Link
S64	2013	FYP	Peer-reviewed journal	*C. Lopez Nozal et al. (2013)*	Moodle FYP management module defining roles, artefacts, and process stages. EMAP: Moodle-based FYP role/artefact model.	Deployment + feedback	Link
S65	2017	FYP	Conference paper	*P. F. M. Khamaruddin et al. (2017)*	FYP workflows integrated into Moodle for multi-stage assessment and progress tracking. EMAP: LMS-integrated milestone tracking.	Deployment + observational study	Link
S66	2018	FYP	Peer-reviewed journal	*V. Sanchez-Anguix et al. (2018)*	Near Pareto-optimal GA for student-supervisor allocation balancing preferences and workload. EMAP: GA-based supervisor allocation algorithm.	Computational experiments + comparison	Link
S67	2024	FYP	Peer-reviewed journal	*H. Yuan et al. (2024)*	Empirical study of student-centred FYP allocation; positive effects on	Empirical study with	Link

					satisfaction and performance. EMAP: preference satisfaction in allocation policy.	performance outcomes	
S68	2016	FYP	Peer-reviewed journal	*O. Abdulkadir et al. (2016)*	GA for allocating project supervisors under preference and workload constraints. EMAP: GA allocation baseline.	Computational evaluation	Link
S69	2022	FYP	Peer-reviewed journal	*D. Manlove et al. (2022)*	Stable-matching algorithms for student-project allocation with preferences. EMAP: stable-matching theoretical foundation.	Algorithmic analysis + experiments	Link
S70	2021	CMS	Conference paper (AAAI)	*A. Cohen et al. (2021)*	Market-inspired bidding scheme for peer-review paper assignment improving fairness. EMAP: bidding-based assignment module design.	Empirical on conference datasets	Link
S71	2023	XTech	Peer-reviewed journal	*Anon. (2023) PLOS ONE*	Survey of COI reporting in open peer review; analyses disclosure practices. EMAP: COI policy and disclosure workflow requirements.	Survey / content analysis	Link
S72	2026	CMS	Platform documentation	*OpenReview (2026)*	Open peer review platform for major AI venues; API-first workflow transparency benchmark. [Platform doc — not empirical.] EMAP: API-first design benchmark.	Platform documentation	Link
S73	2025	JMS	Platform documentation	*Janeway (2025)*	Open-source end-to-end journal platform (submission-to-publication). [Platform doc — not empirical.] EMAP: feature benchmark for publishing pipeline.	Platform documentation	Link

APPENDIX B. FEATURE COMPARISON OF KEY REPRESENTATIVE SYSTEMS

This appendix provides a standardised feature comparison of 20 representative systems from the evidence base with sufficient feature documentation for meaningful cross-system analysis. This appendix is self-contained and fully original to this paper.

TABLE IV — Feature Comparison of Key Representative Systems

ID	System / Study	Yr	Domain	Main Focus	Automation & Analytics	Evaluation Method	Link
J1	Kim et al. (2018) Comparative MMS Analysis	2018	Journal	Compares OJS, ScholarOne, Editorial Manager on workflow, usability, and metadata	Moderate automation; basic reporting	Comparative analysis + expert evaluation	Link
J2	Hasan & Abuelrub (2013) IAJIT OpenConf Usability	2013	Journal	Usability of journal MMS: learnability, efficiency, satisfaction	Basic automation (email, tracking)	Lab usability testing	Link
J3	Mozafari-Vanani et al. (2020) JMS Usability Prioritisation	2020	Journal	Prioritises usability components across multiple JMS	Workflow automated; usability-gap focus	Survey + MCDM	Link
J4	Bogunovic et al. (2003) Early Electronic JMS	2003	Journal	Full-lifecycle JMS: submission, review, publication automation	Full lifecycle automation	Case study at engineering faculty	Link
J5	Kamid et al. (2025) Accreditation-Focused JMS	2025	Journal	JMS for Indonesian university accreditation and KPI evaluation	Automated workflows, KPI dashboards	UAT + accreditation outcomes	Link
J6	Chung et al. (2024) ACOMS+ Open Science	2024	Journal	National open-science system with analytics and API-first design	High; open peer review, analytics API	Case study with feature analysis	Link
C1	ConfiChair (2012) Arapinis et al.	2012	CMS	CMS with formal cryptographic privacy guarantees	Formal access-control protocols	Formal verification + prototype	Link
C2	P3ERS (2012) Aimeur et al.	2012	CMS	Privacy-preserving peer review via cryptography	Cryptographic protocols	Cryptographic analysis	Link
C3	CoCon (2021) Popescu et al.	2021	CMS	Formally verified document confidentiality in CMS	Machine-checked verification	Isabelle/HOL proof + prototype	Link
C4	HotCRP (2008) Kohler	2008	CMS	Large-scale CS conference reviewing with simple workflow	Reviewer assignment, notifications	Experience/design report; wide adoption	Link
C5	TPMS (2013) Charlin & Zemel	2013	CMS	Automated reviewer assignment via topic models	Topic-model similarity matching	Empirical on conference datasets	Link
C6	Reviewerly (2024) Arabzadeh et al.	2024	CMS	IR-based reviewer assignment on large-scale datasets	Information retrieval pipeline	Large-scale empirical IR evaluation	Link

F1	Shabli et al. (2024) FYP + Theory of Shame	2024	FYP	FYP management using Theory of Shame for accountability	Automated notifications, RBAC status tracking	Theory-informed implementation	Link
F2	Isa et al. (2024) FYP Dashboard System	2024	FYP	Dashboard FYP system with RBAC and automated analytics	RBAC dashboards, progress analytics	Prototype + user feedback	Link
F3	Tuah et al. (2022) Smart FYP Dashboard	2022	FYP	Smart dashboard for FYP monitoring with supervisor analytics	Dashboards, institutional data analytics	RAD + user testing	Link
F4	Sanchez-Anguix et al. (2018) GA Supervisor Allocation	2018	FYP	Near Pareto-optimal GA for student-supervisor allocation	Multi-objective genetic algorithm	Computational experiments + comparison	Link
F5	Manlove et al. (2022) Stable Matching Allocation	2022	FYP	Stable-matching algorithms for student-project allocation	Stable matching algorithm variants	Algorithmic analysis + experiments	Link
X1	PEERRec (2024) Bharti et al.	2024	XTech	NLP/AI reviewer recommendation and decision prediction	NLP pipeline, AI triage system	NLP evaluation on ICLR dataset	Link
X2	MRGen (2021) Pradhan et al.	2021	XTech	Deep neural meta-review generation and decision prediction	Deep learning model	DL eval. on ICLR/ACL datasets	Link
X3	Pika et al. (2015) Process Mining	2015	XTech	Bottleneck mining in peer-review systems via event logs	Process mining on structured event logs	Real peer-review event data	Link

REFERENCES

[1] S. Kim et al., "Comparative analysis of manuscript management systems," Science Editing, vol. 5, no. 2, pp. 124-135, 2018. https://doi.org/10.6087/kcse.137

[2] L. Hasan and E. Abuelrub, "Usability testing for IAJIT OpenConf journal management system," Journal of Software, vol. 8, no. 2, pp. 387-396, 2013. https://doi.org/10.4304/jsw.8.2.387-396

[3] A. M. Mozafari-Vanani et al., "Prioritizing usability components in journal management systems," Iranian Journal of Information Processing and Management, vol. 36, no. 2, pp. 443-468, 2020. https://doi.org/10.35050/jipm010.2020.007

[4] H. Bogunovic et al., "An electronic journal management system," in Proc. ITI 2003. https://doi.org/10.1109/ITI.2003.1225350

[5] K. Jacksi, "Design and implementation of online submission and peer review system: Case study of e-journal of University of Zakho," IJSTR, vol. 4, no. 8, pp. 160-165, 2015. http://www.ijstr.org/final-print/aug2015/Design-And-Implementation-Of-Online-Submission-And-Peer-Review-System-A-Case-Study-Of-E-journal-Of-University-Of-Zakho.pdf

[6] K. Kamid et al., "Development and implementation of a web-based journal management system to enhance accreditation," Enrichment: Journal of Multidisciplinary Research, vol. 2, no. 10, 2025. https://doi.org/10.55324/enrichment.v2i10.268

[7] J. Chung et al., "Online submission and review system for open science: ACOMS+," Journal of Information Science Theory and Practice, vol. 12, no. 1, pp. 87-101, 2024. https://doi.org/10.1633/JISTaP.2024.12.1.6

[8] R. M. S. da Fonseca, "A PKP Open Journal System," ResearchGate, 2008. https://www.researchgate.net/publication/37676931

[9] G. Pantaleo and P. Nesi, "Palamede: a multi-press Open Journal System," 2009. https://www.academia.edu/6185629/Palamede_a_Multi_Press_Open_Journal_System

[10] J. A. Teixeira da Silva, "Manuscript management systems require sensible management," Science Editor and Publisher, vol. 10, no. 1, p. 61, 2025. https://doi.org/10.24069/sep-25-37

[11] D. Sidyk-Furman, "An introduction to Open Journal Systems," Academia.edu, 2020. https://www.academia.edu/110708405/An_Introduction_to_Open_Journal_Systems

[12] M. Cyzyk and S. Choudhury, "A survey and evaluation of open-source electronic publishing systems," Open Society Institute, 2008. https://www.academia.edu/18700042/A_survey_and_evaluation_of_open_source_electronic_publishing_systems

[13] B. Owen, "The Public Knowledge Project: Reflections and directions after its first two decades," in Proc. ElPub 2018. https://doi.org/10.4000/proceedings.elpub.2018.18

[14] A. H. M. Shabli et al., "Designing and developing a web-based FYP management system using the Theory of Shame," in Advances in Intelligent Systems and Computing, Springer, 2024. https://doi.org/10.1007/978-3-031-91485-0_20

[15] R. Isa et al., "Prototype development of final year project management system to monitor student performance," Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 40, no. 1, pp. 164-173, 2024. https://doi.org/10.37934/araset.40.1.164173

[16] N. M. Tuah et al., "A dashboard-based system to manage and monitor the progression of undergraduate IT degree FYPs," Pertanika Journal of Science and Technology, vol. 30, no. 1, 2022. https://doi.org/10.47836/pjst.30.1.13

[17] I. T. Afolabi et al., "Decision support system for final year project management," in Proc. WCECS 2019. https://www.iaeng.org/publication/WCECS2019/WCECS2019_pp233-237.pdf

[18] R. Lounas et al., "An e-collaboration application for final-year project management," International Journal of e-Collaboration, vol. 19, no. 1, pp. 1-17, 2023. https://doi.org/10.4018/IJeC.315787

[19] L. Yahaya et al., "Final year students projects allocation and management system," Arid Zone Journal of Basic and Applied Research, vol. 2, no. 2, pp. 83-98, 2023. https://doi.org/10.55639/607

[20] J. V. Joshua et al., "Design and implementation of final year project management system," Data Mining and Knowledge Engineering, vol. 13, no. 6, 2021. https://www.ciitresearch.org/dl/index.php/dmke/article/view/DMKE062021001

[21] S. B. N. Thamrin and T. Andriani, "Design web-based registration and data management of student thesis information system," Jurnal Sains dan Teknologi, vol. 11, no. 1, 2021. https://doi.org/10.30700/jst.v11i1.1111

[22] A. S. Setiabudhi et al., "Frontend design and development of thesis management application using NextJS," Journal of Technology and Science, vol. 8, no. 1, 2025. https://doi.org/10.36378/jtos.v8i1.4385

[23] J. Chung et al., "Open peer review: promoting transparency in open science," Scientometrics, 2020. https://doi.org/10.1007/s11192-020-03488-4

[24] R. Kopak and C.-N. Chiang, "An interactive reading environment for online scholarly journals: The OJS reading tools," OCLC Systems and Services, 2009. https://www.academia.edu/924904/An_interactive_reading_environment_for_online_scholarly_journals_The_Open_Journal_Systems_reading_tools

[25] S. M. Haider and M. Kashif, "Open Journal System," Annals of Abbasi Shaheed Hospital, 2019. https://www.academia.edu/97435296/Open_Journal_System

[26] M. J. Page et al., "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," BMJ, vol. 372, n71, 2021. https://doi.org/10.1136/bmj.n71

[27] K. Petersen et al., "Systematic mapping studies in software engineering," in Proc. EASE 2008. https://doi.org/10.14236/ewic/EASE2008.8

[28] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering (EBSE-2007-01)," Keele University, 2007. https://www.cs.auckland.ac.nz/~norsaremah/2007%20Guidelines%20for%20performing%20SLR%20in%20SE%20v2.3.pdf

[29] J. Webster and R. T. Watson, "Analyzing the past to prepare for the future: writing a literature review," MIS Quarterly, vol. 26, no. 2, pp. xiii-xxiii, 2002. https://aisel.aisnet.org/misq/vol26/iss2/3/

[30] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," MIS Quarterly, vol. 13, no. 3, pp. 319-340, 1989. https://aisel.aisnet.org/misq/vol13/iss3/6/

[31] V. Venkatesh et al., "User acceptance of information technology: Toward a unified view," MIS Quarterly, vol. 27, no. 3, pp. 425-478, 2003. https://aisel.aisnet.org/misq/vol27/iss3/5/

[32] J. Brooke, "SUS: A quick and dirty usability scale," in Usability Evaluation in Industry, Taylor and Francis, pp. 189-194, 1996. https://hci-studies.org/methods-and-measures/downloads/SUS_Brooke1996.pdf

[33] ISO/IEC, "ISO/IEC 25010:2011 - SQuaRE - System and software quality models," 2011. https://www.iso.org/standard/35733.html

[34] G. Arapinis et al., "ConfiChair: A privacy-supporting conference management system," in ESORICS 2012 (LNCS 7459), Springer, pp. 98-115. https://doi.org/10.1007/978-3-642-28641-4_6

[35] E. Aimeur et al., "P3ERS: A privacy-preserving peer-review system," Transactions on Data Privacy, vol. 5, no. 2, pp. 689-712, 2012. https://www.tdp.cat/issues11/tdp.a103a12.pdf

[36] A. Popescu et al., "CoCon: A conference management system with formally verified document confidentiality," Formal Aspects of Computing, vol. 33, pp. 71-109, 2021. https://doi.org/10.1007/s10817-020-09566-9

[37] S. Kalmukov, "Architecture of a conference management system providing advanced paper assignment features," IJCA, vol. 17, no. 8, pp. 30-36, 2011. https://doi.org/10.5120/4083-5888

[38] E. Kohler, "Hot Crap!," in Proc. WOWCS 2008, USENIX. https://www.usenix.org/legacy/event/wowcs08/tech/full_papers/kohler/kohler.pdf

[39] L. Charlin and R. Zemel, "The Toronto Paper Matching System: An automated paper-reviewer assignment system," ICML workshop, 2013. https://www.cs.toronto.edu/~zemel/documents/tpms.pdf

[40] M. Arabzadeh et al., "Reviewerly: Modeling the reviewer assignment task as an information retrieval problem," in Proc. CIKM 2024. https://doi.org/10.1145/3627673.3679081

[41] M. Arabzadeh et al., "exHarmony: Authorship and citations for benchmarking the reviewer assignment problem," arXiv:2502.07683, 2025. https://arxiv.org/abs/2502.07683

[42] S. Shah et al., "Investigating the reviewer assignment problem: A systematic literature review," Journal of Information Science, 2023. https://doi.org/10.1177/01655515231176668

[43] Y. Zhang et al., "A survey and evaluation of reviewer assignment algorithms," Information Processing and Management, vol. 59, no. 6, 2022. https://doi.org/10.1016/j.ipm.2022.103028

[44] "An algorithm for automatic assignment of reviewers to papers," Scientometrics, 2020. doi: 10.1007/s11192-020-03519-0. https://doi.org/10.1007/s11192-020-03519-0

[45] "A data-driven optimization approach for automated reviewer assignment," Intelligent Systems with Applications, 2025. doi: 10.1016/j.iswa.2025.200587. https://doi.org/10.1016/j.iswa.2025.200587

[46] M. Popescu et al., "SmartReviewHub: AI-based scientific conference management system," ICuSi 2024. https://doi.org/10.37789/icusi.2024.6

[47] A. Tenorio-Fornes et al., "Decentralizing science: Towards an interoperable open peer review ecosystem using blockchain," Information Processing and Management, 2021. https://doi.org/10.1016/j.ipm.2021.102724

[48] "Blockchain-Enhanced Peer Review," in Blockchain and Applications (Springer), 2023. doi: 10.1007/978-3-031-66850-0_6. https://doi.org/10.1007/978-3-031-66850-0_6

[49] "Blockchain solutions for scientific paper peer review: A systematic review," Data Technologies and Applications, 2022. doi: 10.1108/DTA-01-2022-0010. https://doi.org/10.1108/DTA-01-2022-0010

[50] "Blockchain and its application in the peer review of scientific works: A systematic mapping review," Publications, 12(4), 40, 2024. https://doi.org/10.3390/publications12040040

[51] "Tokenized Peer Review: An adaptive blockchain mechanism," Journal of Network and Systems Management, 2025. https://doi.org/10.1007/s10922-025-10004-7

[52] "Blockchain-based token system for incentivizing peer review," Decision Support Systems, 2025. https://doi.org/10.1016/j.dss.2025.114514

[53] J. Jin et al., "BeerReview: A blockchain-enabled peer review platform," arXiv:2405.20220, 2024. https://arxiv.org/abs/2405.20220

[54] P. K. Bharti et al., "PEERRec: An AI-based approach to automatically generate recommendations and predict decisions in peer review," International Journal on Digital Libraries, 2024. https://doi.org/10.1007/s00799-023-00375-0

[55] T. Pradhan et al., "A deep neural architecture based meta-review generation and final decision prediction," Neurocomputing, vol. 428, pp. 218-238, 2021. https://doi.org/10.1016/j.neucom.2020.11.004

[56] J. Yu et al., "Automated Peer Reviewing in Paper SEA: Standardization, Evaluation, and Analysis," Findings of EMNLP 2024. https://aclanthology.org/2024.findings-emnlp.595/

[57] G. Russo Latona et al., "The AI Review Lottery: Widespread AI-Assisted Peer Reviews Boost Paper Scores and Acceptance Rates," Proc. ACM, 2024. https://doi.org/10.1145/3757667

[58] K. Kousha and M. Thelwall, "Artificial intelligence to support publishing and peer review: A summary and review," Learned Publishing, 2024. https://doi.org/10.1002/leap.1570

[59] A. Pika et al., "Process modeling and bottleneck mining in online peer-review systems," SpringerPlus, 2015. https://doi.org/10.1186/s40064-015-1183-4

[60] N. Puspaningrum et al., "Development of Review Rating and Reporting in Open Journal System," Procedia Computer Science, 2017. https://doi.org/10.1016/j.procs.2017.10.035

[61] S. Rizqi et al., "Meta Data Extractor Plug-in on OJS to Support Journal Accreditation," in Proc. ICoICT 2023 (Atlantis Press). https://doi.org/10.2991/978-94-6463-364-1_90

[62] M. Liu and A. Samarin, "Cross-site scripting vulnerabilities in Open Journal Systems," International Journal of Electronics and Telecommunications, 2025. https://doi.org/10.24425-ijet.2025.153548

[63] L. Hu and J. Shapiro, "Usability of the submission process in a journal system," in Human-Computer Interaction (Springer), 2015. https://doi.org/10.1007/978-3-319-21383-5_109

[64] C. Lopez Nozal et al., "An innovative Moodle final project management module," IEEE RITA, 2013. https://doi.org/10.1109/RITA.2013.2273109

[65] P. F. M. Khamaruddin et al., "Using Moodle as an integrated FYP management system," in Proc. ICEED 2017. https://doi.org/10.1109/ICEED.2017.8251200

[66] V. Sanchez-Anguix et al., "A near Pareto optimal approach to student-supervisor allocation with two sided preferences," Applied Soft Computing, 2018. https://doi.org/10.1016/j.asoc.2018.11.049

[67] H. Yuan et al., "The impact of a student-centered FYP allocation mechanism on student performance," Humanities and Social Sciences Communications, 2024. https://doi.org/10.1057/s41599-024-03324-7

[68] O. Abdulkadir et al., "A Genetic Algorithm for Allocating Project Supervisors to Students," IJISA, vol. 8, no. 10, 2016. https://doi.org/10.5815/ijisa.2016.10.06

[69] D. Manlove et al., "Student-project allocation with preferences over projects: algorithmic and experimental results," Discrete Applied Mathematics, 2022. https://doi.org/10.1016/j.dam.2020.08.015

[70] A. Cohen et al., "A Market-Inspired Bidding Scheme for Peer Review Paper Assignment," Proc. AAAI, 2021. https://ojs.aaai.org/index.php/AAAI/article/view/16609

[71] "Conflict of interest in the peer review process: A survey," PLOS ONE, 2023. doi: 10.1371/journal.pone.0286908. https://doi.org/10.1371/journal.pone.0286908

[72] OpenReview, "OpenReview: A platform for open peer review (venue documentation)," 2025-2026. https://openreview.net/

[73] Janeway, "Janeway: An open source scholarly publishing platform," 2025. https://janeway.systems/